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Theory(Logic)

This C++ program shows how to use a class to represent a Box and how to overload the + operator so you can add two Boxes together. Inside the Box class three public variables are declared length, breadth, and height — they store the size of the box. setLength(), setBreadth(), and setHeight() are used to set values for the box. getVolume() calculates the volume using the formula: length * breadth * height .The operator+() is used to add two boxes. It takes another Box as input. It creates a new Box and sets its size by adding the corresponding dimensions of both boxes. It prints the memory addresses of the current object (this), the second object (b), and the new object (box). Then it returns the new box. In the main() function: Three boxes (Box1, Box2, and Box3) are created. Their addresses in memory are printed. Sizes are set for Box1 and Box2. Their volumes are calculated and printed. Box3 = Box1 + Box2 creates a new box with combined dimensions. It prints the dimensions and volume of all three boxes.

Pseudo Code

- **Step 1: Include input-output library**
- **Step 2: Use standard namespace**
- **Step 3: Create class named Box**
 - **Public:**
 - **Variables: length, breadth, height**
 - **Function getVolume()**
 - **return length * breadth * height**
 - **Function setLength(len)**
 - **set length = len**
 - **Function setBreadth(bre)**
 - **set breadth = bre**
 - **Function setHeight(hei)**
 - **set height = hei**
 - **Overload operator+ with another box b**
 - **Create new Box called box**
 - **box.length = this.length + b.length**
 - **box.breadth = this.breadth + b.breadth**
 - **box.height = this.height + b.height**
 - **print address of box**
 - **print address of b**
 - **print address of this (current object)**
 - **return box**
- **Step 4: Start main function**
 - **Create Box1, Box2, Box3**
 - **Print addresses of Box1, Box2, and Box3**
 - **Set length, breadth, and height for Box1**
 - **Set length, breadth, and height for Box2**
 - **Calculate and print volume of Box1**
 - **Calculate and print volume of Box2**
 - **Box3 = Box1 + Box2**
 - **Print lengths, breadths, and heights of all three boxes**
 - **Calculate and print volume of Box3**

- **Step 5:End main function**

Program Code

```
#include<iostream>
using namespace std;
class Box
{
    public:
    double getVolume(void)
    {
        return length*breadth*height;
    }
    void setLength(double len)
    {
        length=len;
    }
    void setBreadth(double bre)
    {
        breadth=bre;
    }
    void setHeight(double hei)
    {
        height=hei;
    }

    Box operator+(const Box& b)
    {
        Box box;
        box.length=this->length+b.length;
        box.breadth=this->breadth+b.breadth;
        box.height=this->height+b.height;
        cout<<"\nAddress of Box is"<<&box;
        cout<<"\nAddress of b is"<<&b;
        cout<<"\nAddress of this is"<<this;
        return box;
    }
}
```

```

    }
    double length;
    double breadth;
    double height;
};

int main()
{
    Box Box1;
    Box Box2;
    Box Box3;
    double volume=0.0;
    cout<<"\nAddress of Box1 is"<<&Box1;
    cout<<"\nAddress of Box2 is"<<&Box2;
    cout<<"\nAddress of Box3 is"<<&Box3;

    Box1.setLength(6.0);
    Box1.setBreadth(7.0);
    Box1.setHeight(5.0);

    Box2.setLength(12.0);
    Box2.setBreadth(13.0);
    Box2.setHeight(10.0);

    volume=Box1.getVolume();
    cout<<"\nvolume of Box1:"<<volume<<endl;

    volume=Box2.getVolume();
    cout<<"\nvolume of Box2:"<<volume<<endl;

    Box3=Box1+Box2;
    cout<<"\n\nBox1 length is "<<Box1.length;
    cout<<"\nBox1 breadth is "<<Box1.breadth;
    cout<<"\nBox1 height is "<<Box1.height;

    cout<<"\n\nBox2 length is "<<Box2.length;

```

```
cout<<"\nBox2 breadth is "<<Box2.breadth;
cout<<"\nBox2 height is "<<Box2.height;

cout<<"\n\nBox3 length is "<<Box3.length;
cout<<"\nBox3 breadth is "<<Box3.breadth;
cout<<"\nBox3 height is "<<Box3.height;

volume=Box3.getVolume();
cout<<"\nVolume of Box3:"<<volume<<endl;
return 0;
}
```

Output

```
Address of Box1 is 0x7fff6c5ab7a0
Address of Box2 is 0x7fff6c5ab7c0
Address of Box3 is 0x7fff6c5ab7e0
volume of Box1: 210

volume of Box2: 1560

Address of Box is 0x7fff6c5ab770
Address of b is 0x7fff6c5ab7c0
Address of this is 0x7fff6c5ab7a0

Box1 length is 6
Box1 breadth is 7
Box1 height is 5

Box2 length is 12
Box2 breadth is 13
Box2 height is 10

Box3 length is 18
Box3 breadth is 20
Box3 height is 15
Volume of Box3: 5400
```


Conclusion

Code is run successfully, required output is received.